MSE 2 · CASE STUDY PRESENTATION · AY 2026-27

FROM WASTE STREAM
TO VALUE LOOP

E-Waste Management, Circular Economy & Recycling Models

Analysis of Global Best Practices, Indian Regulations, City Hubs & Corporate Strategies

PRESENTER [Student Name] ROLL NO. [Roll No.] CLASS / DIV. [Class / Div]

Ten case studies. One circular ambition.

This deck traces the end-to-end e-waste chain — from product design, through reduction, reuse and segregation, into formal recycling and material recovery — using ten cases spanning global frameworks, Indian policy, municipal models, the informal economy and corporate circularity.

GUIDING PRINCIPLE

Design → Reduce / Reuse → Segregation → Formal Recycling → Material Recovery

CORE THEMES

01

Extended Producer Responsibility

EPR mandates and legal mechanisms making producers accountable for end-of-life recovery.

02

Informal Workforce Integration

Bringing waste pickers, aggregators and dismantlers into formal recovery networks.

03

Closed-Loop Product Design

Modularity, disassembly and recycled inputs designed in at the manufacturing stage.



03 · GLOBAL BENCHMARK

JAPAN

Discipline as infrastructure.

Municipal discipline, standardized source separation and intermediate volume reduction turn everyday citizens into the first stage of the recovery chain.

38.11M

tonnes

Municipal waste, FY2024

839g

per person / day

Waste generation rate

99.2%

pre-disposal

Waste reduction rate

41.9%

of 991 sites

Waste-to-energy incineration

LESSONS — Ward-level segregation · transparent public data · enforced collection schedules

04 · GLOBAL BENCHMARK

SWITZERLAND

Financing the return loop.

An Advance Recycling Fee (ARF) is levied at purchase and administered by two Producer Responsibility Organizations — SENS eRecycling and SWICO — that fund frictionless consumer drop-off nationwide.

132K

tonnes

WEEE processed 2023 (+9% YoY)

10.36

kg / resident

Handled by SENS across 509 hubs

1,381

companies

Manufacturers & importers financing ARF

Predictable financing plus zero-cost consumer return equals the highest formal channelization in Europe.



India’s E-Waste Rules 2022.

Effective 1 April 2023 · MoEFCC & CPCB · Expanded scope: 106 EEE items across 7 sectors

01
Producers & Importers
Register on the centralized digital portal for EEE placed on market.

02
EPR Recycling Targets
Binding recycling-credit obligations issued against annual sales.

03
Registered Recyclers
Process material and issue certificates through the CPCB portal.

04
Refurbishing Route
Refurbishing certificates incentivize extended product life cycles.

05
Audit & Penalty
Third-party audits and environmental penalties enforce compliance.

KEY MECHANISM · Certificate trading on the CPCB portal converts producer obligations into a transparent digital marketplace, replacing earlier collection-target compliance.

Real impact vs. paper compliance.

India’s 2022 rules transition producers from collection targets to binding recycling-credit obligations. The question is whether verified physical recovery keeps pace with the paperwork.

STRENGTHS

- Measurable annual recycling quotas
- Producer accountability at scale
- Centralized digital oversight
- Tradable certificate marketplace

BOTTLENECKS

- Risk of paper-only compliance
- Unverified credit trading
- Recycling capacity shortages
- Weak informal-sector linkage

STRATEGIC IMPERATIVES

- Risk-based physical audits of units
- Link certificates to batch-level recovery
- Formalize informal aggregator chains
- Public dashboards for mass balance

THE TECHNOLOGY HUB

Bengaluru



India’s largest IT corridor drives one of the fastest hardware turnover cycles in the country — and the largest concentration of corporate obsolescence.

CHALLENGES

Fragmented informal dismantling · hazardous processing in unregulated units · data-security risk on discarded corporate drives

SOLUTIONS

Chain-of-custody for IT handovers · bulk campus collection drives · certified refurbishment centers linked to registered recyclers

12,000

tonnes / year · Computer e-waste baseline

30–40K

corporate PCs · Obsolete annually in the city

Mumbai’s density is its blueprint.

A high-density commercial hub with long-standing informal scrap markets — where e-waste routinely mixes into the daily municipal stream.

31,622
tonnes / year · E-waste estimate for Mumbai

47,469
tonnes / year · MMR region estimate (2012)

6,200
tonnes / day · MSW absorbing mixed e-waste

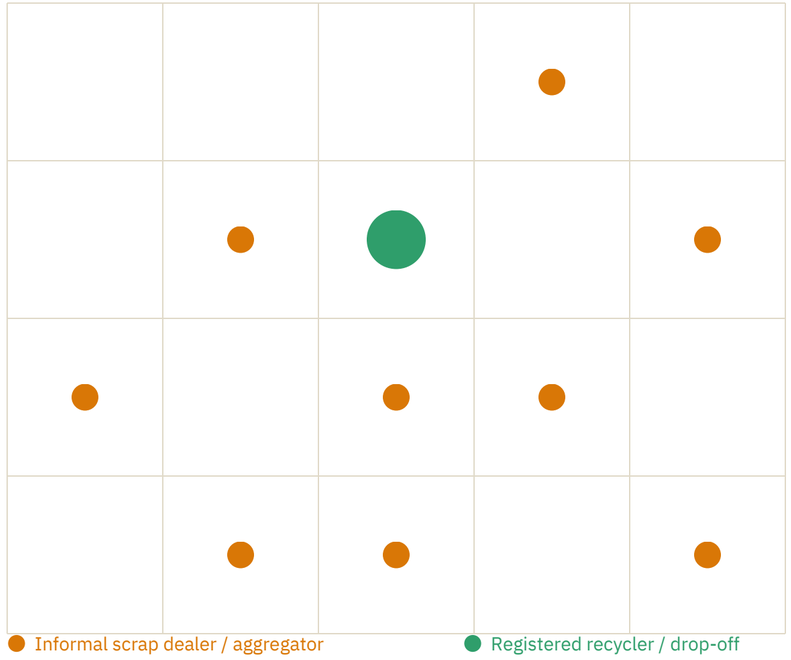
INTEGRATION ROADMAP

01 Ward-wise mapping of scrap dealers and aggregators

02 Formal buy-back contracts with registered recyclers

03 Community drop-off centers diverting from municipal waste

SCHEMATIC · WARD-WISE INFORMAL HUBS





09 · DECENTRALIZED MODEL

CLEAN KERALA COMPANY LIMITED (CKCL)

A state that collects at scale.

State-backed coordination links local governance with authorized recyclers. Households hand dry waste to 35,352 Haritha Karma Sena members; CKCL aggregates, sorts and routes hazardous streams.

7.71M

households

Covered by the collection network

2,678

MT e-waste

Collected by CKCL to date

205

MT hazardous

E-waste routed for safe processing

35,352

workers

Haritha Karma Sena members

Cooperatives as civic infrastructure.

An autonomous waste-picker cooperative partnered with the Pune Municipal Corporation. Training, PPE and financial incentives turn pickers into the city's frontline e-waste collectors.

3,900+
waste pickers
Integrated across ~10 lakh properties

1,500
tonnes / day
Collected daily (1.5–1.6K range)

~35%
diversion
220 t/day diverted from landfill

100K
MT CO_2
Emissions avoided annually



A hidden circular economy at scale.



A dense cluster of 300–500 sorting and trading units employing 4,000–6,000 workers moves 500–800 tonnes of plastic and materials every day — roughly 187,000 tonnes a year.

REALITIES

High recovery rates and deep market expertise coexist with environmental and social risks that formalization must address, not erase.

POLICY IMPLICATION

Formalize with upskilling, PPE and direct off-take agreements linking clusters to registered recyclers.

Designing waste out of the product.

An upstream strategy built on circular product design, modularity and closed-loop material recovery — turning the corporate producer into an active recycler.

CORE PILLARS

01
Durability

02
Dematerialization

03
Design for Disassembly

04
Asset Recovery

2030 MILESTONES

1.1B kg

Used electronics recovered since 2007 (~2.5 billion lbs).

97.6%

Packaging recycled or renewable — targeting 100% by 2030.

1 : 1

2030 goal: reuse or recycle one tonne for every tonne sold; >50% recycled content in products.

Six models, one lens.

MODEL / ENTITY	CORE MECHANISM	KEY STRENGTH	MAJOR CHALLENGE
Japan	Municipal discipline & intermediate reduction	99.2% diversion / reduction rate	High infrastructure capital & opex
Switzerland	Advance Recycling Fee + PROs (SENS / SWICO)	Stable financing & convenient return	Battery & composite device complexity
India EPR 2022	Centralized digital portal & tradable credits	Scalable producer accountability	Verifying physical recovery vs. paper
Kerala · CKCL	State aggregation + Haritha Karma Sena	Doorstep reach across millions of homes	Downstream processing capacity limits
Pune · SWaCH	Worker-owned cooperative partnership	Livelihood security + landfill diversion	Scrap price volatility
Dell	Closed-loop circular engineering	Upstream waste prevention & recycled inputs	Complex global supply-chain tracing

Policy, people and product design — as one ecosystem.

Sustainable e-waste management demands policy enforcement, inclusive social infrastructure and upstream circular design — unified into a single ecosystem.

01

Holistic Integration

Link informal collectors to CPCB-registered recyclers through structured aggregation incentives and traceable off-take contracts.

02

Upstream Incentives

Expand public procurement mandates favoring modular, repairable and refurbished electronics — pulling circular design into the mainstream.

03

Data Integrity

Shift from collection-receipt tracking to physical, batch-level mass-balance audits verified by third parties on the CPCB portal.